¿Qué es un “modelo”?





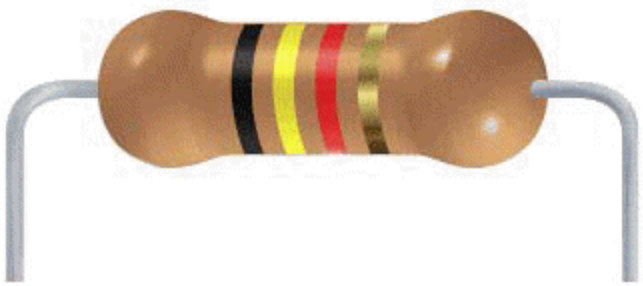


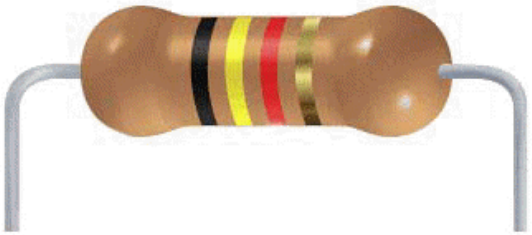
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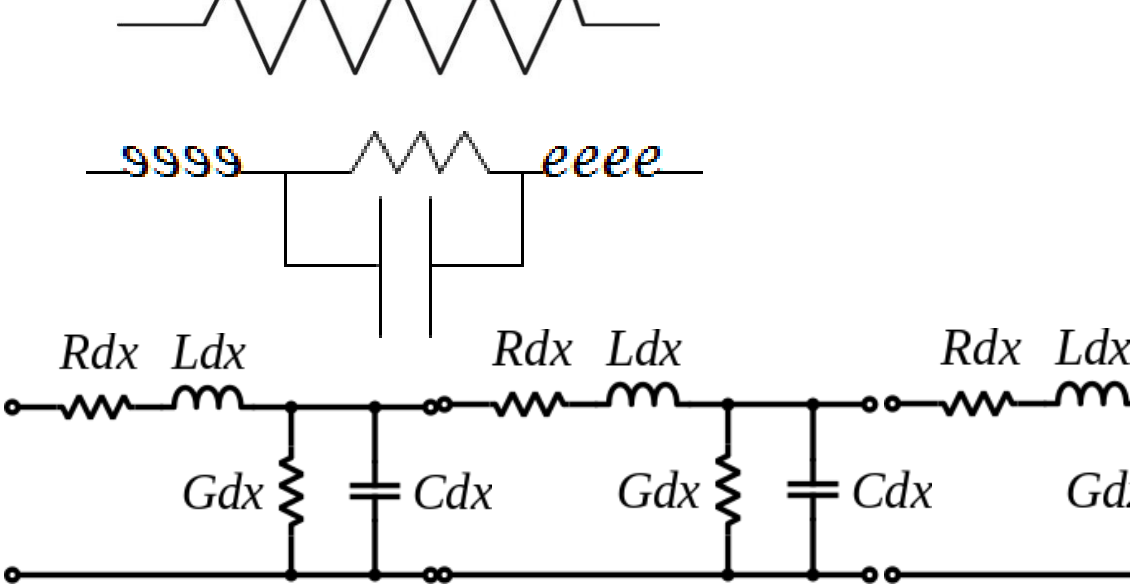
¿Es un “resistor” o una “resistencia”?

“resistencia” del “resistor”





Frec.↑
ó
L→λ



$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

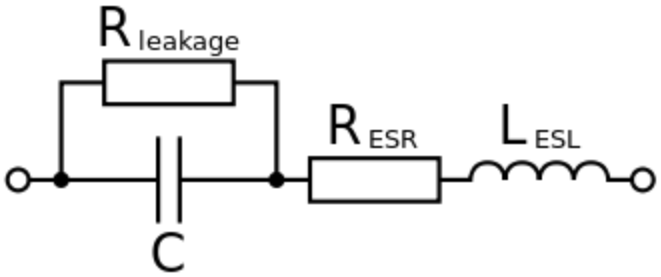
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

El rango de validez del modelo
= f (tecnología, f, T, V, H%, etc.)











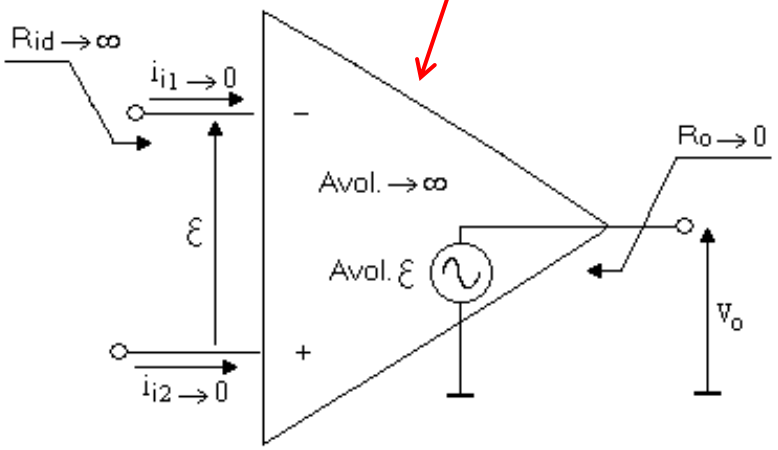


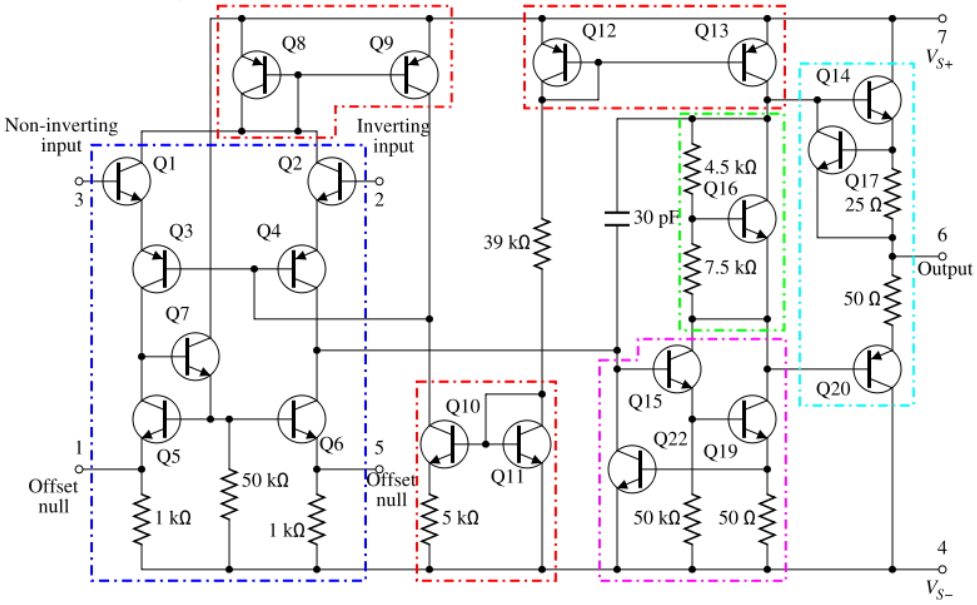
1KHz Frec. 1Mhz

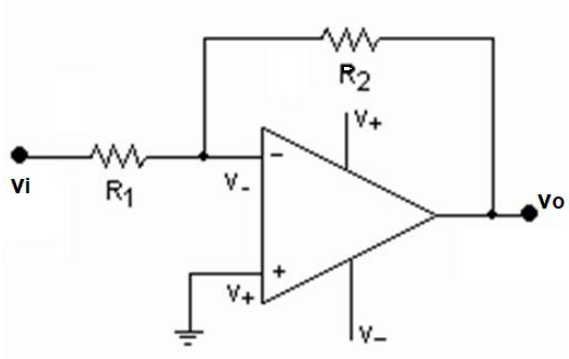


Modelo de un amplificador operacional

macromodelo



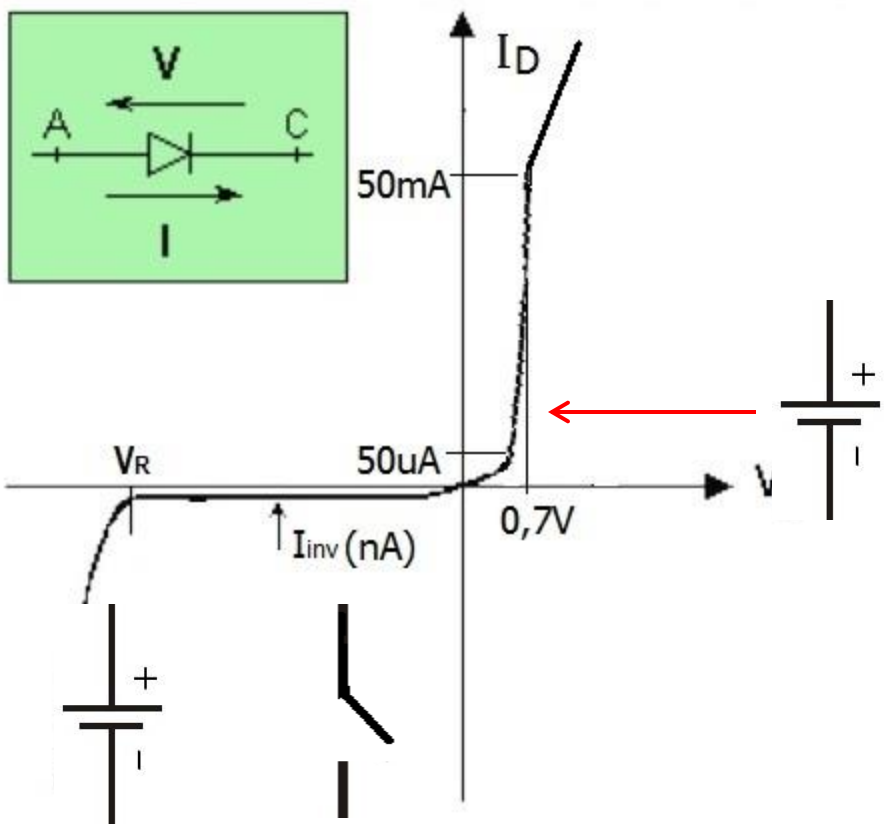




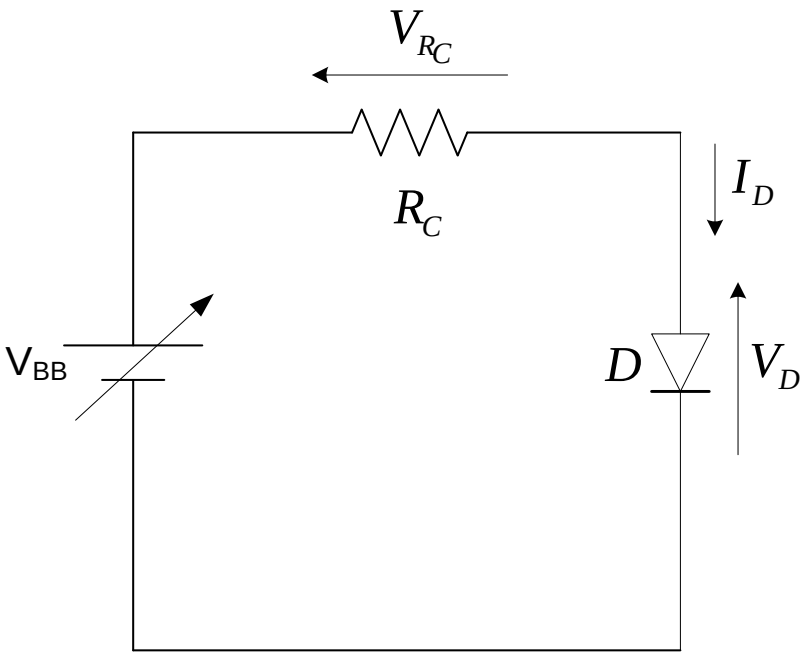
$v_o/v_i = - R_2 / R_1$

Si es válido el modelo del OPAMP ideal.

Modelo de un diodo (elemento **no lineal**)

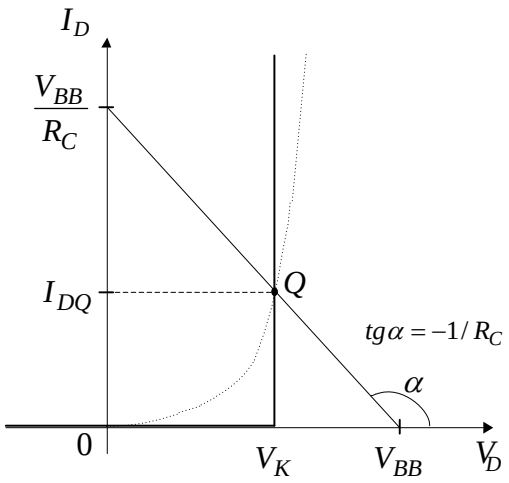


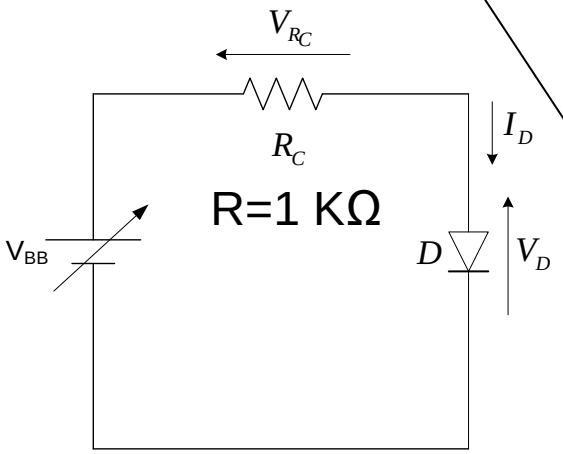
Característica estática de un diodo de Si de bajo nivel de potencia



Diodo ideal: $$I_D = I_S.\left(e^{V_D/V_T} - 1\right) \cong I_S.\, e^{V_D/V_T}$$

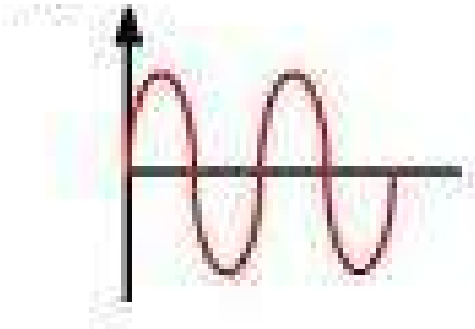
El modelo básico o “0” sirve
para muchas aplicaciones **(pero no siempre)**

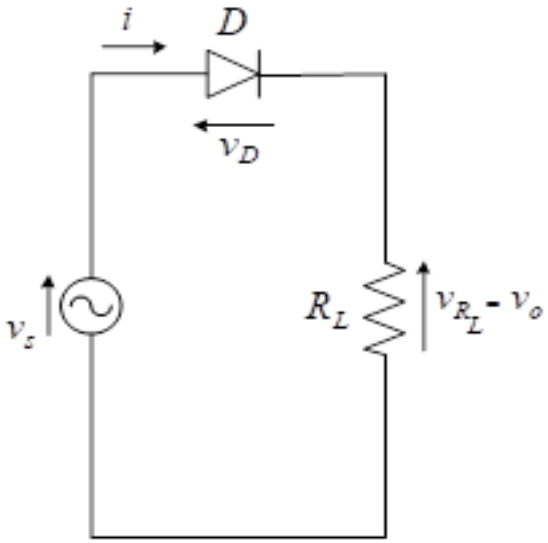


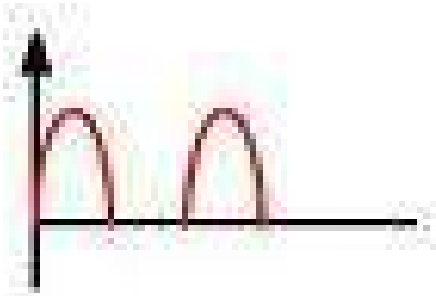


Si ¿R=1 MΩ?

Cómo se comporta el diodo ante una senoidal de **gran amplitud**?



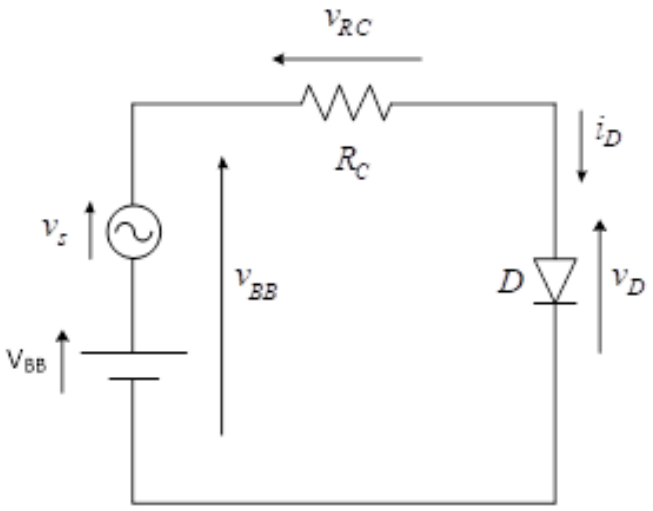


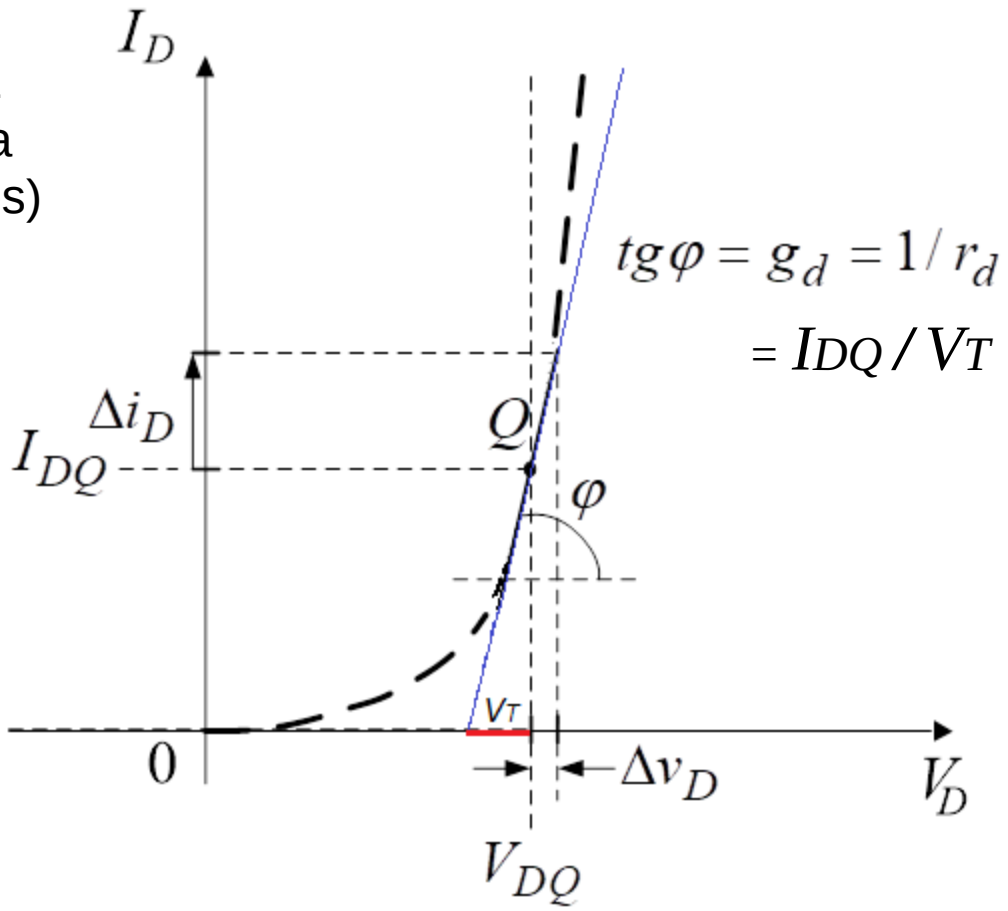


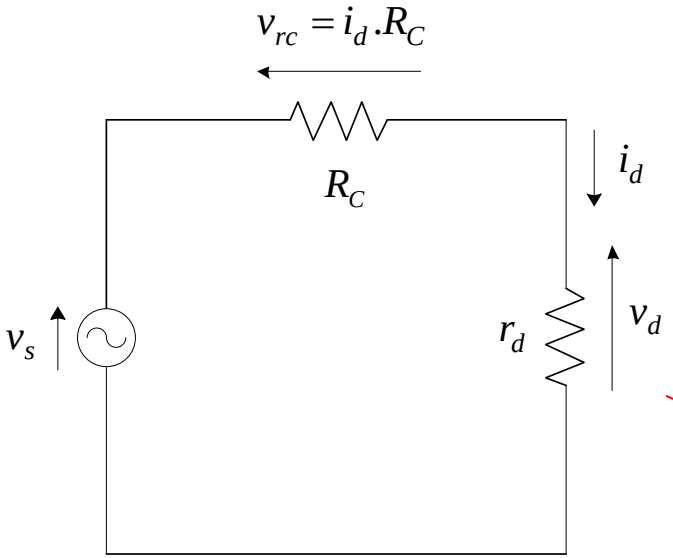
Señal rectificada

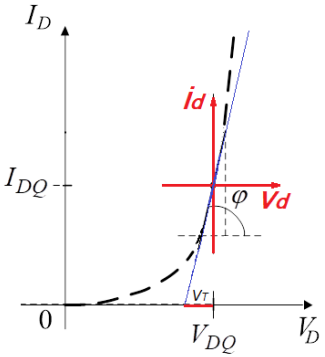
Pequeña señal:
(condición para poder aplicar las herramientas
de resolución de **sistemas lineales**)

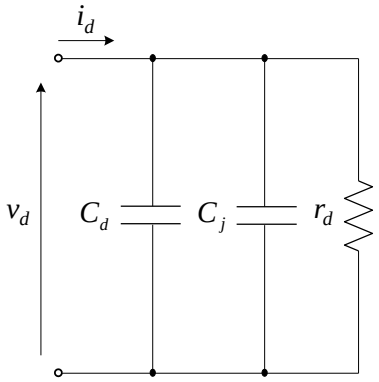
Ejemplo del diodo en directa
(igual procedimiento en inversa
o ruptura pero con otros valores)











Y si tenemos en cuenta los efectos reactivos en el modelo lineal.

En resumen…

Reposo:

$$V_{BB} = I_{DQ} R_C + V_{DQ}$$

Pequeña señal:

$$v_s = i_d \, . R_C + i_d \, . r_d$$

Valores totales:

$$v_D = V_{DQ} + v_d \cong V_{DQ} + \hat{V}_d \, senwt \qquad i_D = I_{DQ} + i_d \cong I_{DQ} + \hat{I}_d \, senwt$$

No se está aplicando el “principio de superposición”